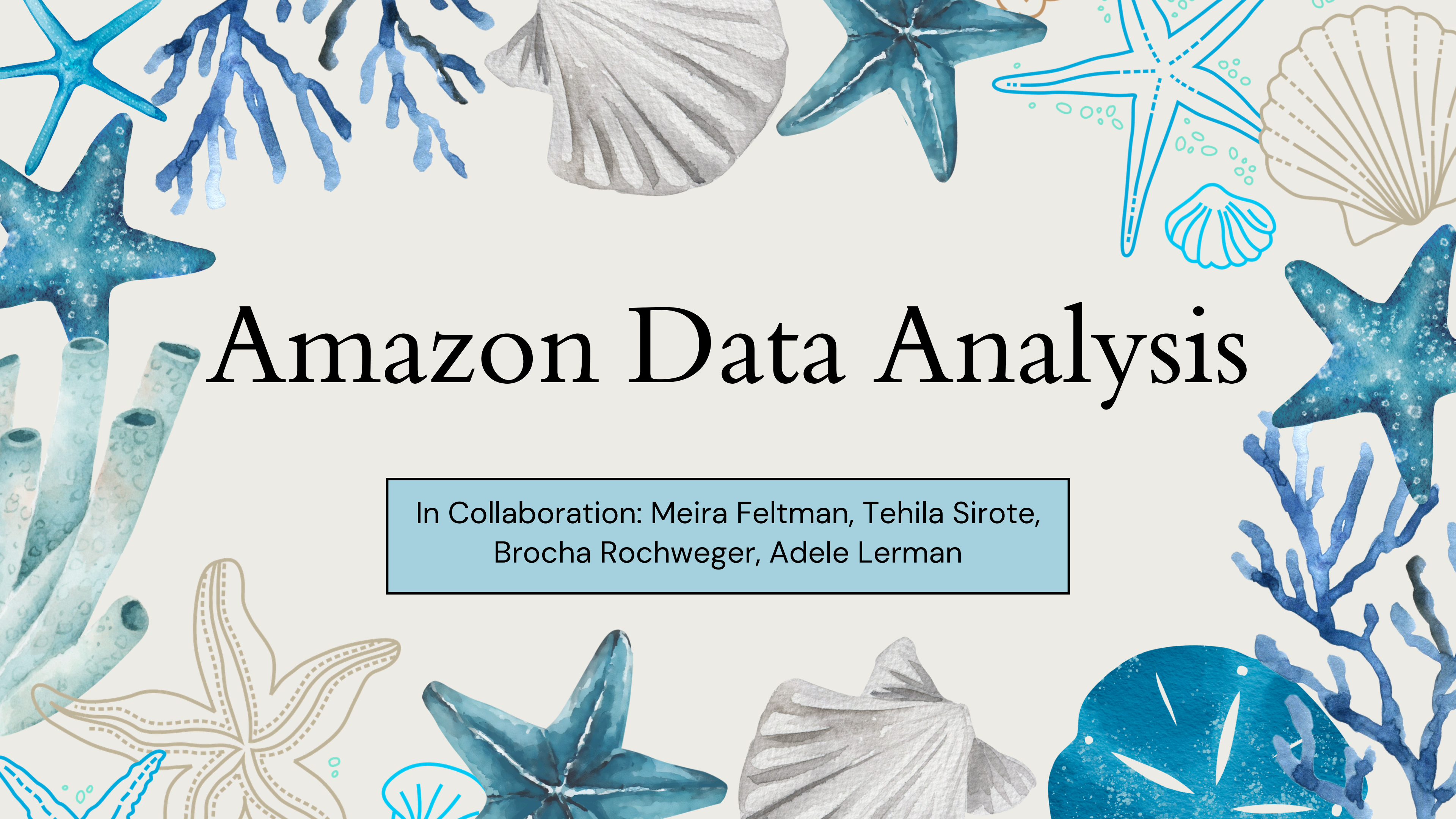




Amazon Data Analysis

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Analytics Contents

- 01 Data Summary
- 02 Classification logistic regression and k n (supervised)
- 03 Regression (supervised)
- 04 K mean Clustering (unsupervised)
- 05 Association (unsupervised)
- 06 Conclusion

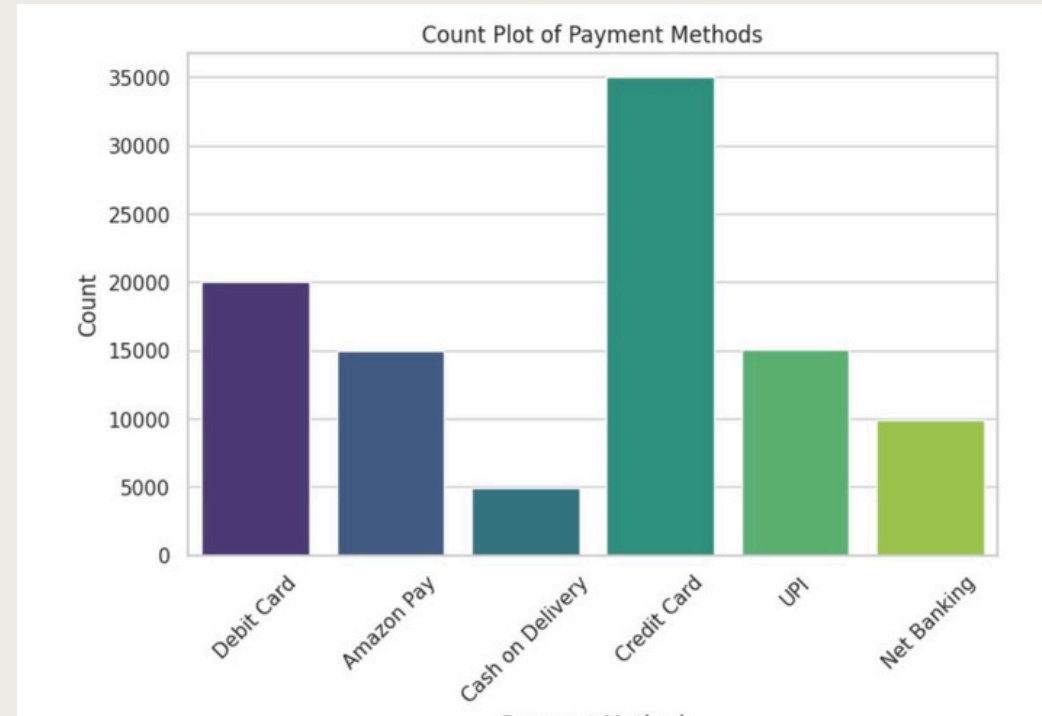
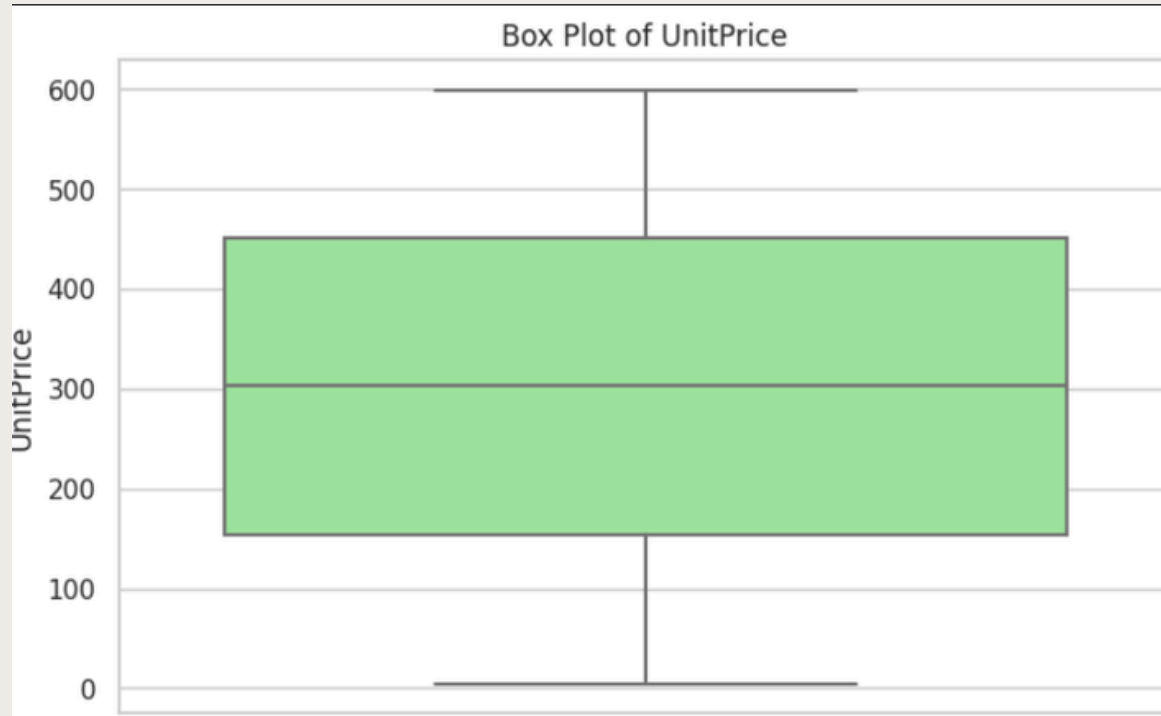


About Dataset

Synthetic Amazon-style e-commerce sales data for analytics and ML

This dataset contains 100,000 synthetic Amazon-style e-commerce sales transactions, designed to closely resemble real-world online retail behavior. With 20 clean and well-structured columns, it captures detailed information about customers, products, pricing, payments, logistics, and order outcomes. (Kaggle, Rohit Kumar)

EDA

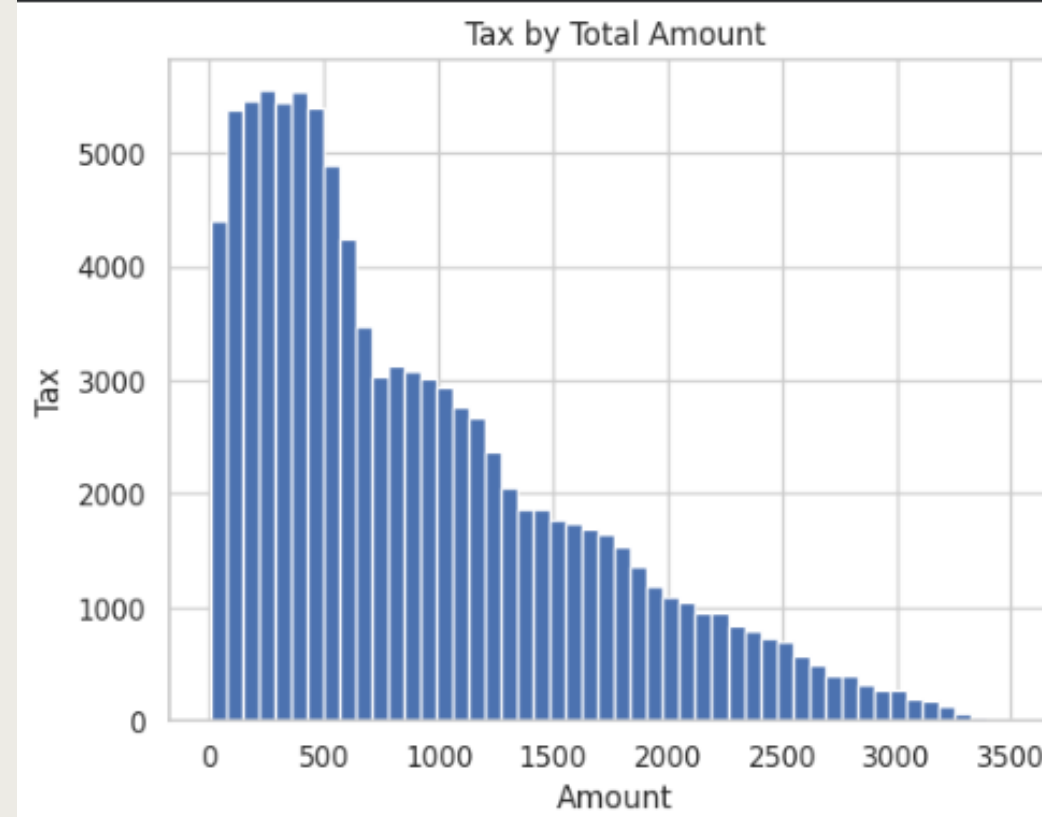
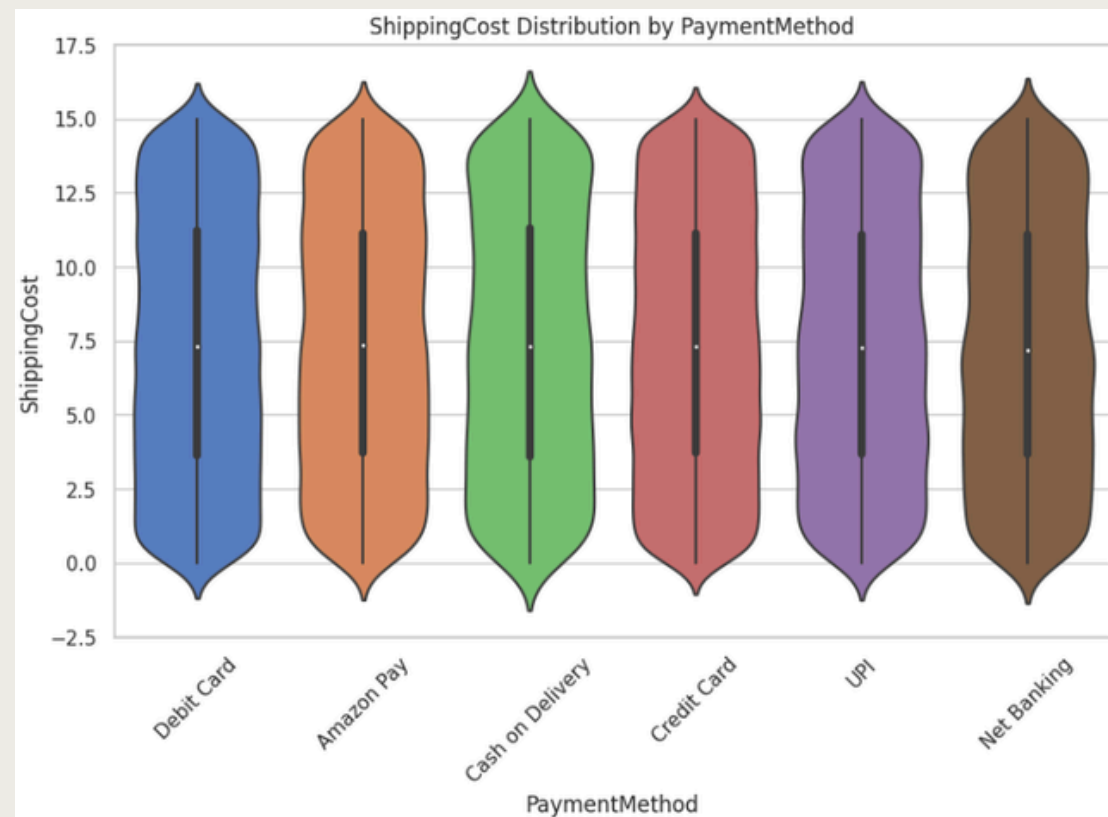


	OrderID	OrderDate	CustomerID	CustomerName	ProductID	\
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	

	ProductName	Category	Brand	Quantity	UnitPrice	\
0	Drone Mini	Books	BrightLux	3	106.59	
1	Microphone	Home & Kitchen	UrbanStyle	1	251.37	
2	Power Bank 20000mAh	Clothing	UrbanStyle	3	35.03	
3	Webcam Full HD	Home & Kitchen	Zenith	5	33.58	
4	T-Shirt	Clothing	KiddoFun	2	515.64	

	Discount	Tax	ShippingCost	TotalAmount	PaymentMethod	OrderStatus	\
0	0.00	0.00	0.09	319.86	Debit Card	Delivered	
1	0.05	19.10	1.74	259.64	Amazon Pay	Delivered	
2	0.10	7.57	5.91	108.06	Debit Card	Delivered	
3	0.15	11.42	5.53	159.66	Cash on Delivery	Delivered	
4	0.25	38.67	9.23	821.36	Credit Card	Cancelled	

	City	State	Country	SellerID
0	Washington	DC	India	SELL01967
1	Fort Worth	TX	United States	SELL01298
2	Austin	TX	United States	SELL00908
3	Charlotte	NC	India	SELL01164
4	San Antonio	TX	Canada	SELL01411





Regression

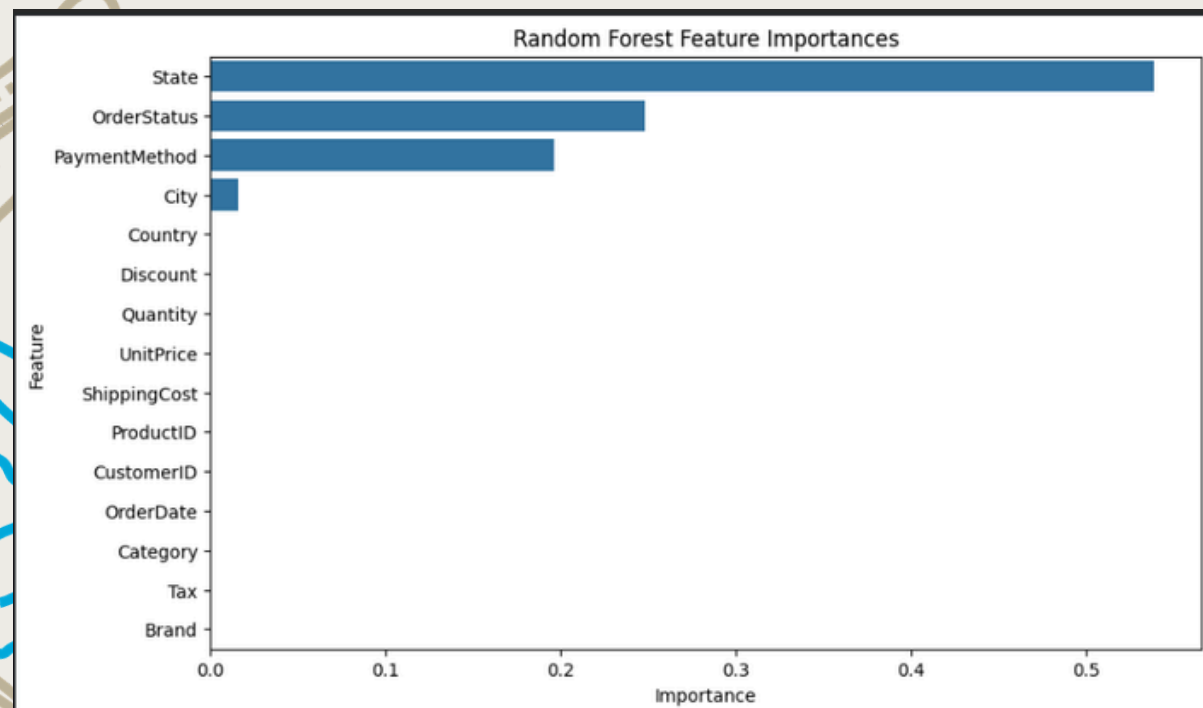
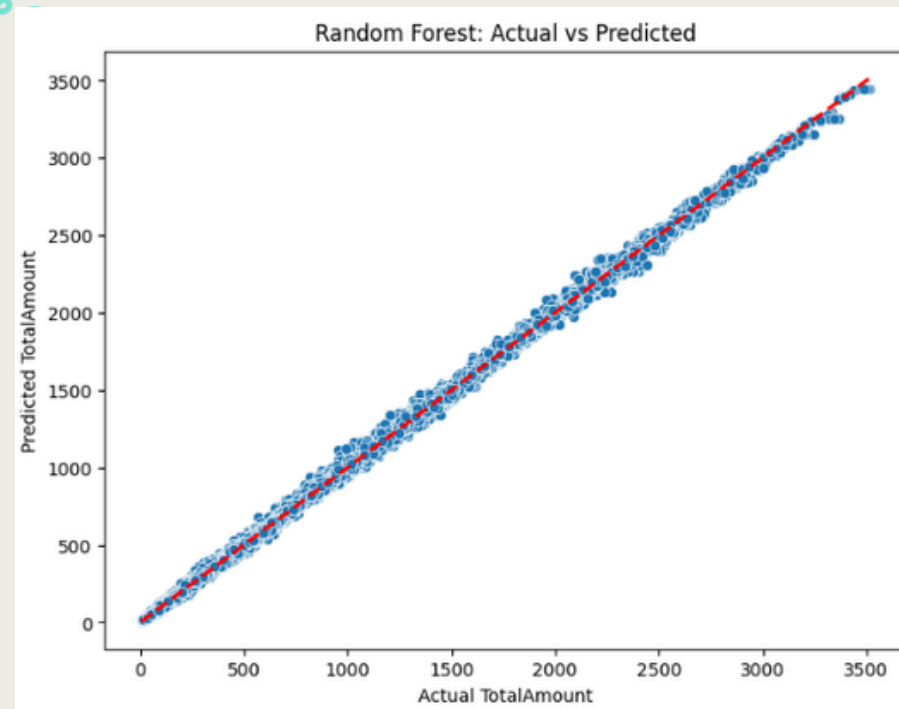
Random Forest– Supervised

Objective: This model builds 100 different decision trees using random samples of the data, and the Random Forest predicts an order's total by averaging the predictions from all trees, which greatly improves accuracy and stability.

Results: The Random Forest model accurately predicted Amazon order totals with minimal error and explained nearly all customer spending behavior.

Regression

Random Forest– Supervised



RMSE ≈ 21 On average, the model predicts TotalAmount with an error of about \$21. If an order is really \$350, the model might predict around \$329 or \$371.

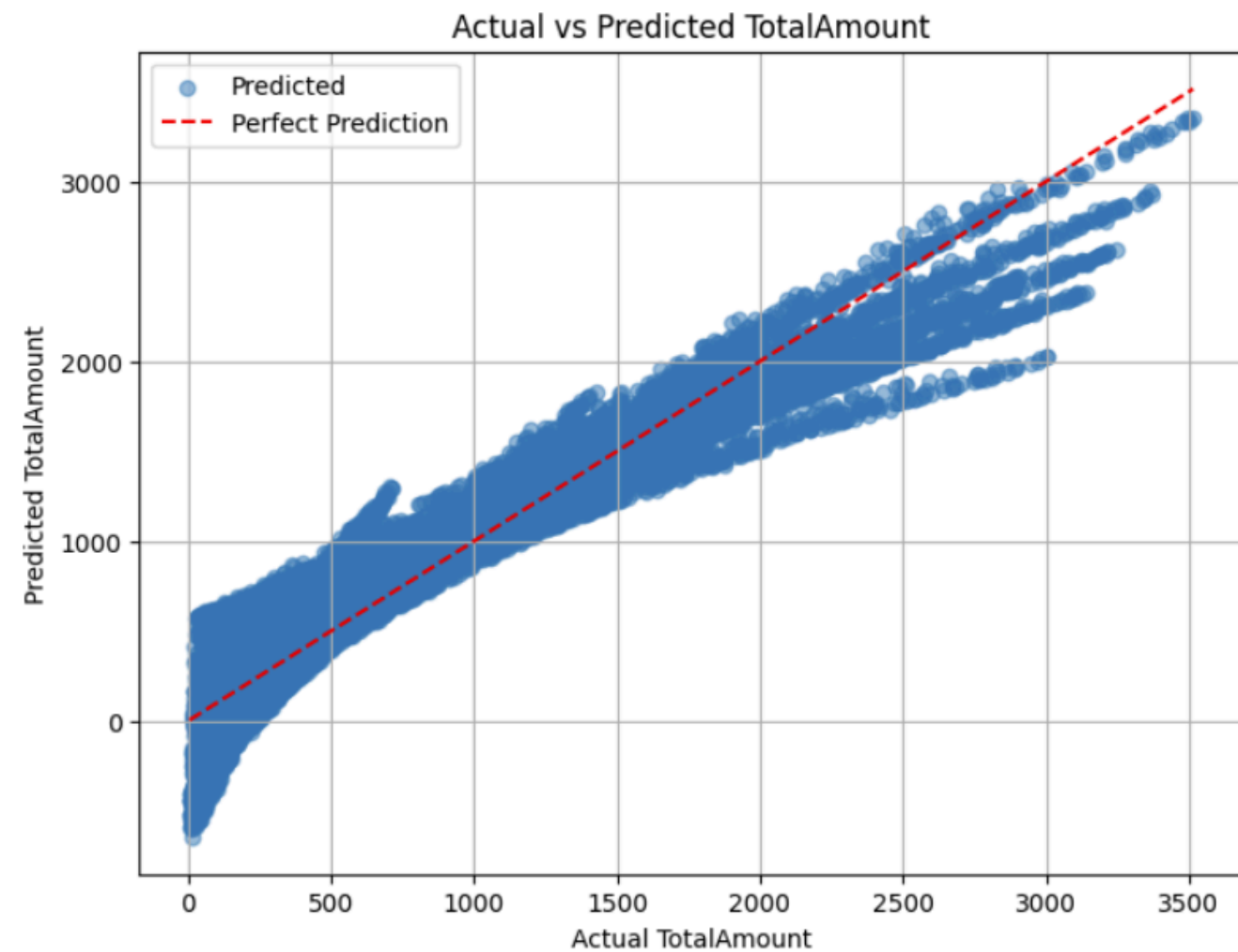
MAE ≈ 15 The average mistake is \$15

$R^2 = 0.9992$. The model explains 99.92% of the patterns in the Amazon sales.

Cross-Validation $R^2 = 0.9991$. The model is consistent, meaning it performs almost the same on different parts of the dataset

Classification

Logistic Regression – Supervised



The goal of logistic regression is to predict the probability that an input belongs to a particular class using labeled training data.

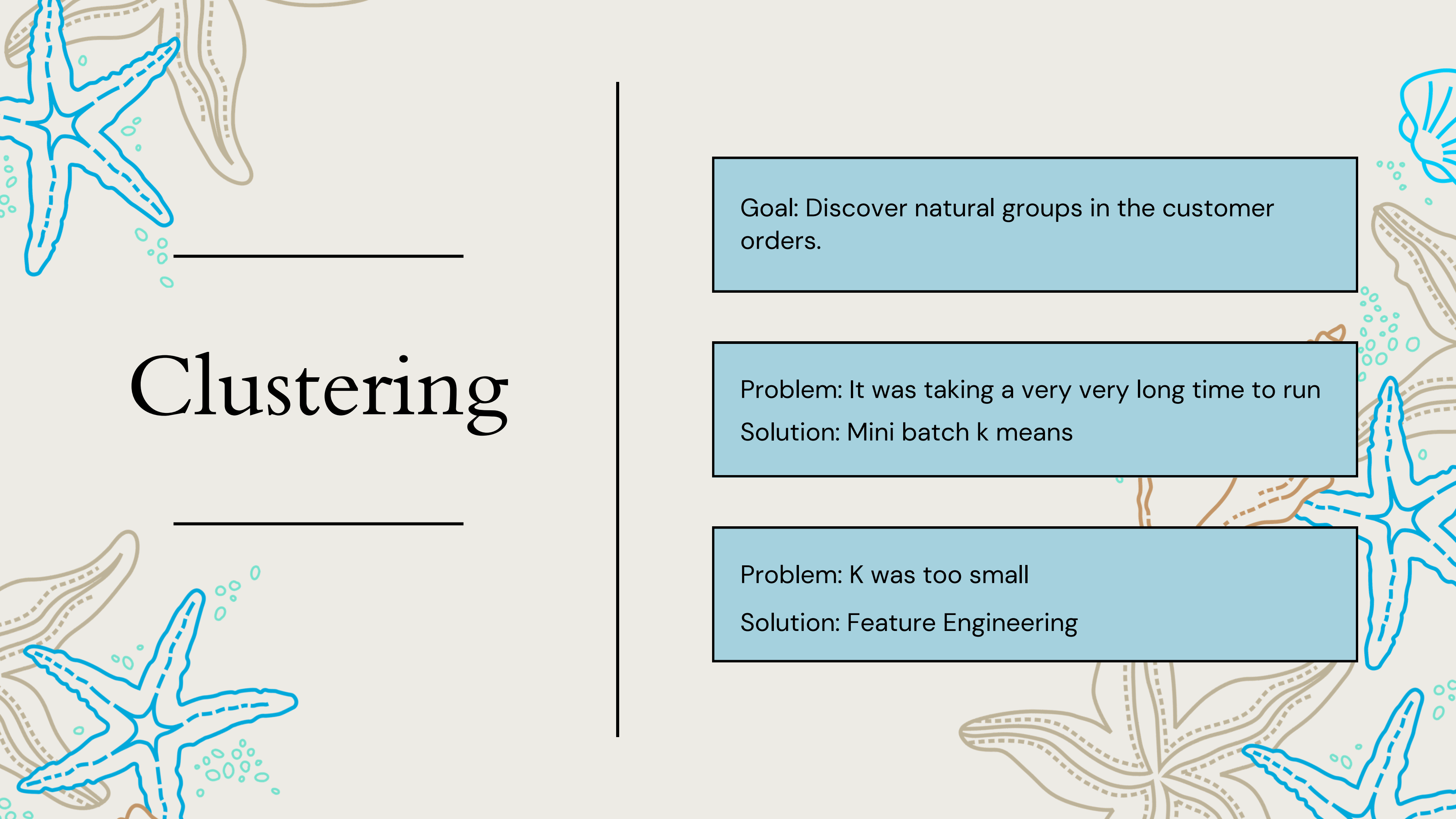
It learns a decision boundary that separates classes by modeling the relationship between features and a binary or categorical outcome.

RMSE $\approx$ 217 On average, the model predicts TotalAmount with an error of about 217. If the total amount is \$500 so the model can error and predict \$717 or \$383.

R^2 Score = 0.9093, model explains about 91% of the changes in order totalamounts.

This Model is predicting the total amount based on brand, category, payment method and a few other features. Each Blue Dot represents one prediction. The red line shows where the prediction and actual were the same so the prediction was correct.

By the Lower amounts the model seems to be performing more accurately (compared to higher amounts) and as it gets higher the model has less correct predictions.



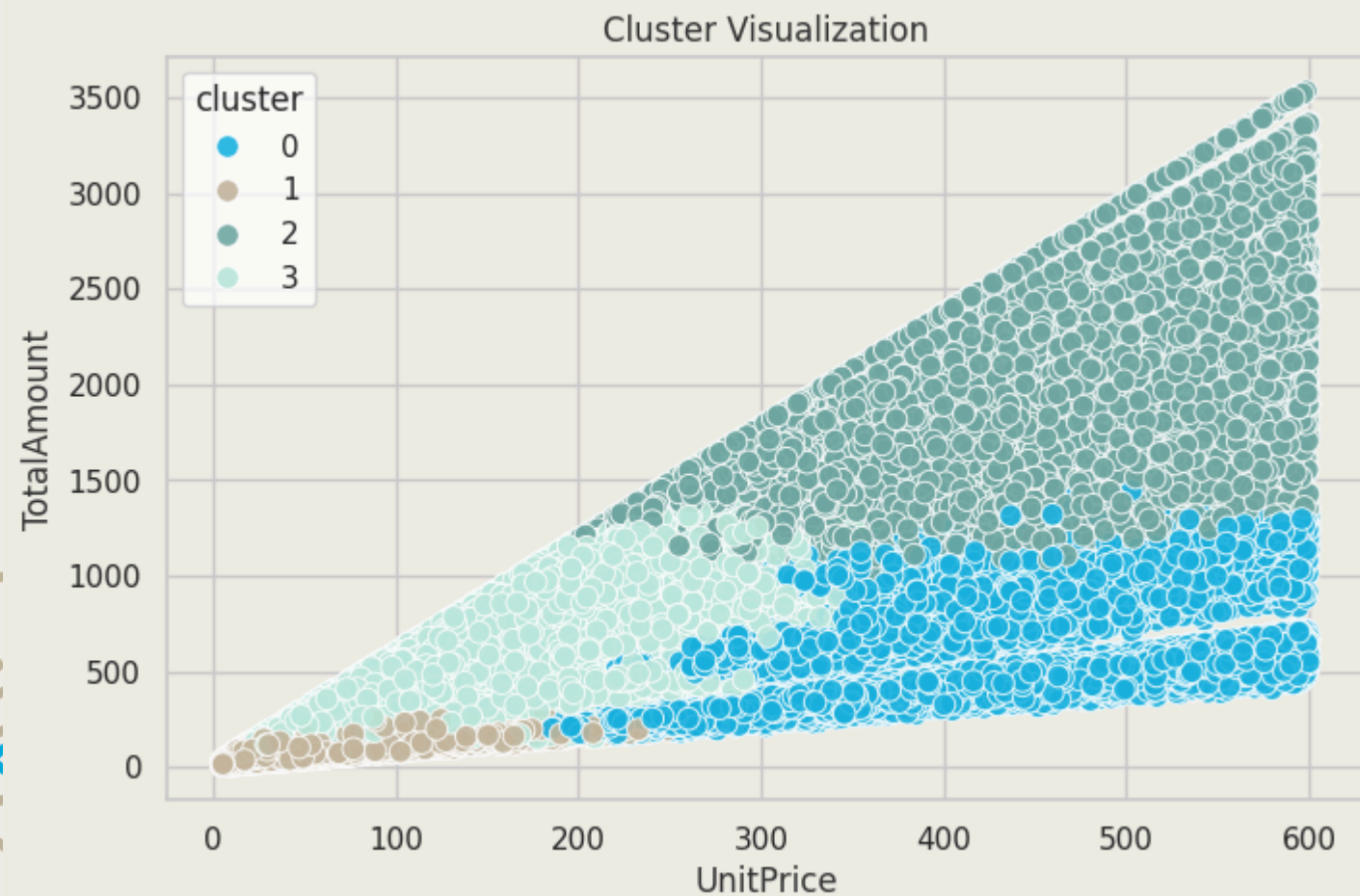
Clustering

Goal: Discover natural groups in the customer orders.

Problem: It was taking a very very long time to run
Solution: Mini batch k means

Problem: K was too small
Solution: Feature Engineering

Clustering RESULTS



Group 0: Premium Shoppers – buy single expensive items, no discounts and low shipping impact.

Group 1: Budget Shoppers– buy low cost items with small orders. Shipping makes up a bigger part of their purchase.

Group 2: Bulk Shoppers–make large orders of semi-expensive items, they are the highest spenders.

Group 3: Mid Range– buy regularly priced things, in regular quantities.

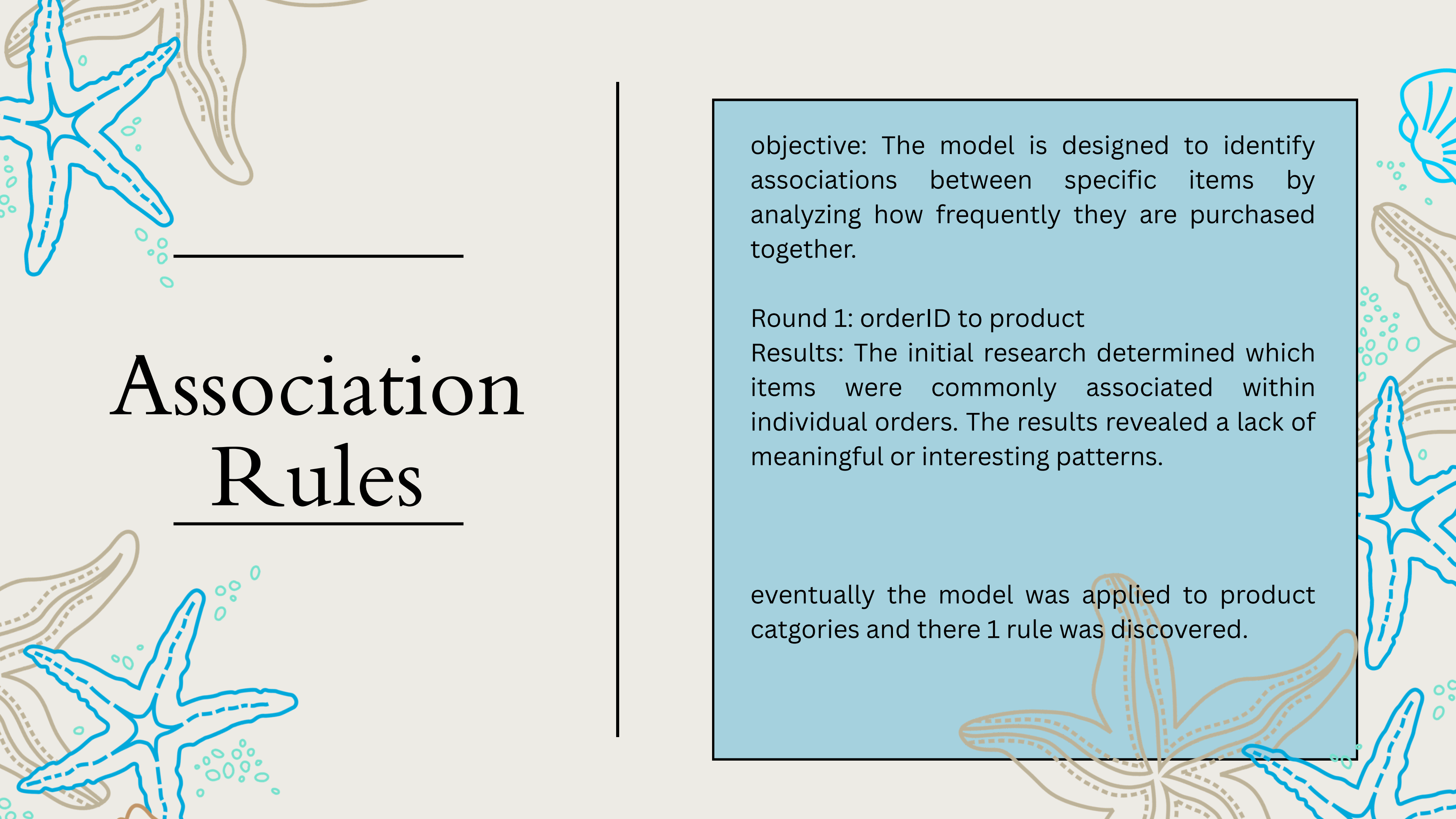
Association rules method:

The model groups your rows by each (CustomerID, ProductName) pair. it then adds up the total Quantity that customer bought of that product across all their orders. it then converts quantities to binary presence/absence:

If customer bought at least 1 $\rightarrow$ 1

Otherwise $\rightarrow$ 0

Apriori scans your basket matrix and finds item sets (products) that appear together often enough.



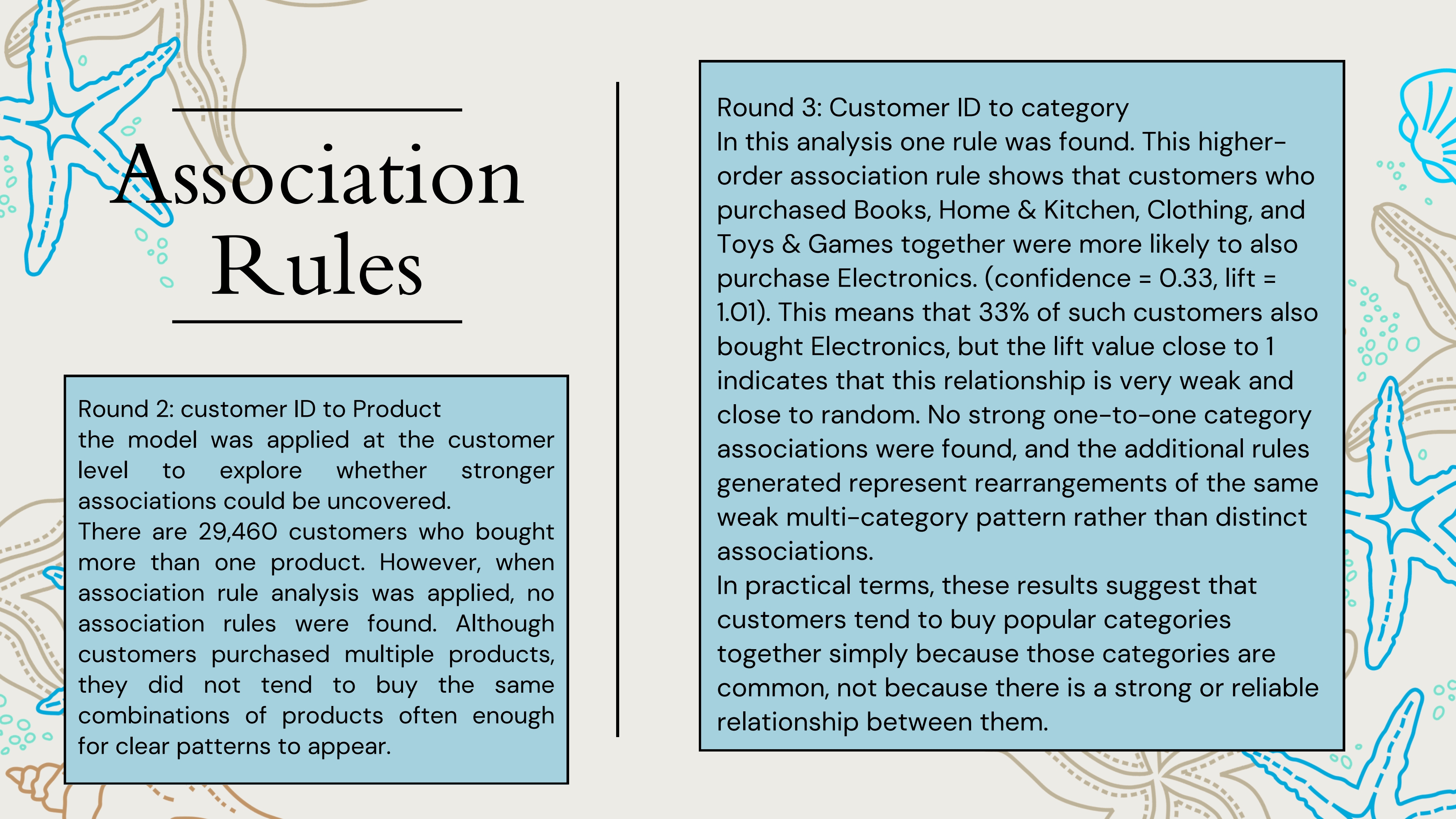
Association Rules

objective: The model is designed to identify associations between specific items by analyzing how frequently they are purchased together.

Round 1: orderID to product

Results: The initial research determined which items were commonly associated within individual orders. The results revealed a lack of meaningful or interesting patterns.

eventually the model was applied to product categories and there 1 rule was discovered.



Association Rules

Round 2: customer ID to Product
the model was applied at the customer level to explore whether stronger associations could be uncovered.

There are 29,460 customers who bought more than one product. However, when association rule analysis was applied, no association rules were found. Although customers purchased multiple products, they did not tend to buy the same combinations of products often enough for clear patterns to appear.

Round 3: Customer ID to category

In this analysis one rule was found. This higher-order association rule shows that customers who purchased Books, Home & Kitchen, Clothing, and Toys & Games together were more likely to also purchase Electronics. (confidence = 0.33, lift = 1.01). This means that 33% of such customers also bought Electronics, but the lift value close to 1 indicates that this relationship is very weak and close to random. No strong one-to-one category associations were found, and the additional rules generated represent rearrangements of the same weak multi-category pattern rather than distinct associations.

In practical terms, these results suggest that customers tend to buy popular categories together simply because those categories are common, not because there is a strong or reliable relationship between them.

Conclusion

We used four data mining models to study Amazon-style sales data. The classification model helped predict outcomes using labeled data but was less accurate than other methods. The regression model performed very well and predicted order totals with very small error. Clustering grouped customers into clear types, such as high spenders, budget shoppers, and average buyers. The association model showed that customers do not consistently buy the same items together, and most patterns were due to popular categories rather than strong product relationships. Together, these models gave a clear and complete picture of customer buying behavior.

References

AI assistance was employed in parts of the research process.

The End

